

Vaibhav Kumar Mahawar

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Ajmer, Rajasthan, India

Education

Dr. B. R. Ambedkar National Institute of Technology Jalandhar

Master of Technology in Machine Intelligence and Automation CGPA: 8.11/10

Jalandhar, Punjab

Aug. 2024 – Jun. 2026

Poornima Institute of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering CGPA: 8.54/10

Jaipur, Rajasthan

Jul. 2019 – Jun. 2023

Experience

Software Development Intern

Jul. 2022 – Sep. 2022

Techienest Pvt. Ltd.

Jaipur, Rajasthan

- Developed a React Native mobile application serving 500+ students by integrating REST APIs, progress tracking, and interactive learning modules while improving application usability.
- Collaborated with cross-functional teams to develop, test, and optimize scalable mobile application features while integrating REST APIs and improving application performance and maintainability.

Projects

RudraAI– Intelligent Software Engineering Platform | *Node.js, Typescript, Python, Redis, Neo4j, Docker* 2026

- Developed a scalable software engineering platform using FastAPI, Node.js, Redis, Neo4j, and Docker to automate repository analysis, code generation, debugging, documentation, and developer workflows.
- Designed scalable backend services with FastAPI, Redis, and Neo4j while implementing memory-enabled reasoning, tool execution, and intelligent agent collaboration for software engineering workflows.

Career Twin– Full-Stack Career Platform | *React, Typescript, AWS Bedrock, LangGraph, Vector Search* 2026

- Developed a full-stack career platform using React, TypeScript, FastAPI, and AWS Bedrock for resume analysis, job matching, and interview preparation.
- Built scalable backend APIs integrating vector search, semantic retrieval, and document processing to deliver personalized career recommendations.

Drone Surveillance Analytics Platform | *Python, YOLOv8, OpenCV, LangChain, LLMs, ANFIS, RAG* 2026

- Developed a modular surveillance analytics platform integrating computer vision inference pipelines with backend processing for drone detection and threat analysis.
- Implemented backend services for surveillance summarization, contextual analytics, and natural language interaction over detection results.

Research Experience

Real-Time Drone Detection and Classification System | *Python, YOLOv8, OpenCV, ANFIS, Fuzzy Logic* 2026

- Developed a real-time YOLOv8-based drone detection framework with ANFIS-enhanced classification, achieving over 91% detection accuracy while reducing bird–drone misclassification by 52%.

Generative AI and Agentic Systems Research | *LLMs, LangChain, LangGraph, MCP, RAG, AI Agents* Ongoing

- Exploring autonomous AI agents, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), long-context reasoning, and multi-agent orchestration for enterprise-scale AI applications.

Technical Skills

Programming Languages: Python, Java, C++, JavaScript, TypeScript, SQL

AI & Machine Learning: Scikit-learn, OpenCV, YOLO, Computer Vision, CNNs, Transformers, Vision Transformers, OpenAI, Gemini, LangChain, LangGraph, RAG, MCP, AI Agents, Embeddings, Vector Search

Backend Development: FastAPI, REST APIs, Node.js, Express.js

Frontend Development: React.js, React Native, HTML, CSS

Databases: Neo4j, Redis, PostgreSQL, MongoDB, SQLite, FAISS, Weaviate

Cloud & Devops AWS, Docker, Kubernetes, Git, GitHub, GitHub Actions, Linux, Azure

Tools: VS Code, Eclipse, Jupyter Notebook, Postman, Linux

Certifications & Achievements

Research Publications Published research papers in international journals and conferences on Computer Vision, Artificial Intelligence, Intelligent Surveillance, Photovoltaic Systems, and Renewable Energy Optimization.

Microsoft Certified: Azure AI Fundamentals (AI-900) Certified in Azure AI services, Machine Learning fundamentals, Responsible AI, and AI solution development.

GATE Qualified Qualified GATE Computer Science (2024 & 2026) and GATE Data Science & Artificial Intelligence (DA 2026).